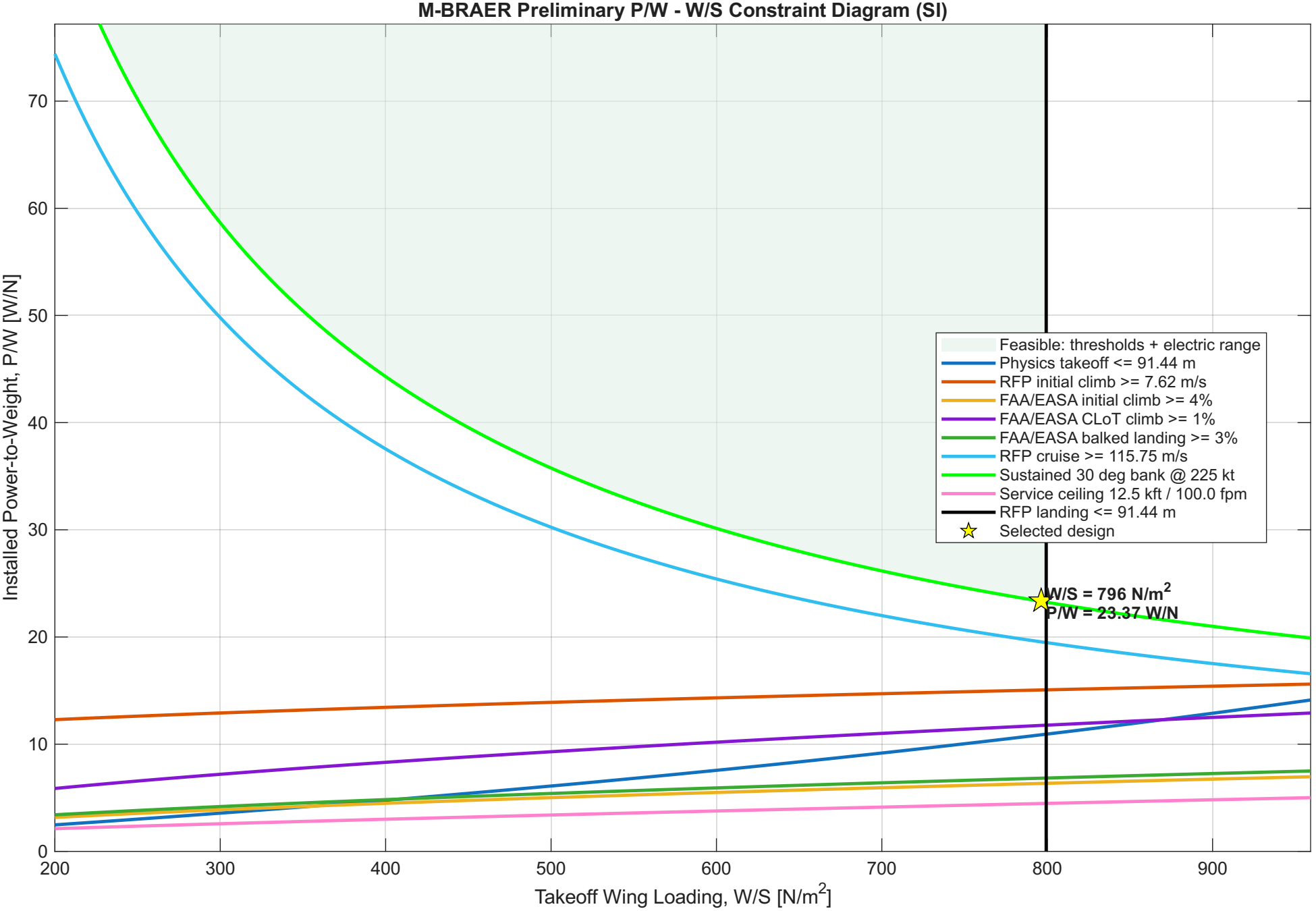

M-BRAER Preliminary P/W - W/S Constraint Diagram (SI)
Installed Power-to-Weight, P/W [W/N]
Takeoff Wing Loading, W/S [N/m^2]
0
10
20
30
40
50
60
70
200
300
400
500
600
700
800
900
Feasible: thresholds + electric range
Physics takeoff <= 91.44 m
RFP initial climb >= 7.62 m/s
FAA/EASA initial climb >= 4%
FAA/EASA CLoT climb >= 1%
FAA/EASA balked landing >= 3%
RFP cruise >= 115.75 m/s
Sustained 30 deg bank @ 225 kt
Service ceiling 12.5 kft / 100.0 fpm
RFP landing <= 91.44 m
Selected design
W/S = 796 N/m^2
P/W = 23.37 W/N